

D778

Advanced Software Engineering

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A medical-device company is building firmware for an insulin pump. Regulators fix the requirements, full documentation is mandatory, and any change needs formal approval. The team has 25 engineers in two time zones. Which process model fits best?

- A. A plan-driven Waterfall model.
- B. Code-and-fix with testing at the end.
- C. Split the work into one team per time zone.
- D. Scrum with requirements discovered each sprint.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which project is a better fit for Agile than for Waterfall?

- A. A consumer mobile app whose features shift with weekly user feedback.
- B. Flight-control software under strict certification.
- C. A payroll system built to a fixed government specification.
- D. Any project with more than 10 people.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A requirements defect would cost \$200 to fix during requirements review. Using the rule of thumb that fix cost rises about 10× per later phase (requirements → design → coding → testing), what might it cost if first found in testing? The team uses a modern bug tracker.

- A. \$800
- B. \$200,000
- C. \$20,000
- D. \$2,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A team completed 30, 34, and 26 story points in its last three sprints. The remaining backlog is 180 points. The team recently moved to a new office. About how many sprints will it take to finish?

- A. About 5.3 sprints
- B. 2 sprints
- C. About 6.9 sprints
- D. 6 sprints

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A team's average velocity is 40 points. The release is in 5 sprints. How many backlog points can the Product Owner plan into the release?

- A. 40
- B. 200
- C. 8
- D. 45

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A hospital's scheduling software passes every unit and integration test against its specification. In user acceptance testing, schedulers find it cannot handle split shifts, which they use daily. The hospital has 900 staff. What failed?

- A. Unit testing.
- B. Scaling — 900 staff is too many.
- C. Verification — the product does not match its specification.
- D. Validation — the right product was not built, though it was built to spec.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

In the V-model, which test level checks the architectural (high-level) design?

- A. Acceptance testing.
- B. Unit testing.
- C. System testing.
- D. Integration testing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

At a fintech company with three Scrum teams, a senior developer and the Scrum Master disagree about whether feature A or feature B should come first in the product backlog. The developer has 12 years of experience. Who decides?

- A. The Product Owner.
- B. The senior developer, based on experience.
- C. The Scrum Master.
- D. The team votes, and the majority always wins.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A contractor is starting a first-of-its-kind system with major unknowns in its sensor technology. Each cycle must identify the biggest risks and reduce them with prototypes before committing further. The budget is \$40 million. Which model fits?

- A. Big-bang: build everything, then test.
- B. Spiral.
- C. Scrum.
- D. Waterfall.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

To make a trade-show date, a team hard-coded prices and skipped tests in the checkout module. Six months later, every change there takes three times as long. The trade show drew 5,000 visitors. What is this, and how should it be handled?

- A. Rewrite the whole system from scratch right away.
- B. Scope creep.
- C. Technical debt; plan steady refactoring and add tests to the checkout module.
- D. Plan another trade show to recover costs.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A late project has 5 developers. The manager adds 5 more, all strong coders from a top training program. How many communication paths are there now, compared with before?

- A.** 90 now vs. 20 before.
- B.** 100 now vs. 25 before.
- C.** 45 now vs. 10 before.
- D.** 10 now vs. 5 before.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A team has 28 communication paths. How many people are on the team?

- A.** 7
- B.** 14
- C.** 8
- D.** 28

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A medical-device company is building firmware for an insulin pump. Regulators fix the requirements, full documentation is mandatory, and any change needs formal approval. ~~The team has 25 engineers in two time zones.~~ Which process model fits best?

Target word: process model **Decides:** Stable, regulated, documentation-heavy requirements.

A	Correct	Fits fixed requirements, formal sign-offs, and heavy documentation.
B	Adjacent category	Lacks the discipline regulated work demands.
C	Camouflage	Team layout does not decide the process model.
D	Inverted operation	Built for changing requirements, the opposite of this case.

2. Answer A Concept 1 · transfer

Which project is a better fit for Agile than for Waterfall?

Target word: Agile fit — reversed **Decides:** Requirements change with feedback.

A	Correct	Frequent change favors short iterations.
B	Confusion set	Regulated and documentation-heavy.
C	Inverted operation	Stable requirements fit plan-driven work.
D	Absolute language	Team size alone does not decide.

3. Answer B Concept 2

A requirements defect would cost \$200 to fix during requirements review. Using the rule of thumb that fix cost rises about 10× per later phase (requirements → design → coding → testing), what might it cost if first found in testing? ~~The team uses a modern bug tracker.~~

Target word: late fix cost **Decides:** Three phases later: $\times 10 \times 10 \times 10$.

A	Adjacent category	Grows linearly ($\$200 \times 4$), not by a factor per phase.
B	Correct	$\$200 \times 1,000 = \$200,000$.
C	Wrong stem number	Stops at coding, one phase short.
D	Dropped element	Applies only one phase of growth.

4. Answer D Concept 3

A team completed 30, 34, and 26 story points in its last three sprints. The remaining backlog is 180 points. ~~The team recently moved to a new office.~~ About how many sprints will it take to finish?

Target word: forecast sprints **Decides:** Average velocity = $90 / 3 = 30$.

A	Adjacent category	Uses the best sprint (34).
B	Dropped element	Divides by the three-sprint total, dropping the average.
C	Confusion set	Uses the worst sprint (26).
D	Correct	$180 / 30 = 6$.

5. Answer B Concept 3 · transfer

A team's average velocity is 40 points. The release is in 5 sprints. How many backlog points can the Product Owner plan into the release?

Target word: solve for capacity **Decides:** Capacity = velocity × sprints.

A	Dropped element	One sprint only.
B	Correct	$40 \times 5 = 200$.
C	Inverted operation	Divides velocity by sprints.
D	Wrong stem number	Adds the two numbers.

6. Answer D Concept 4

A hospital's scheduling software passes every unit and integration test against its specification. In user acceptance testing, schedulers find it cannot handle split shifts, which they use daily. ~~The hospital has 900 staff.~~ What failed?

Target word: V&V classification **Decides:** Built to spec, but the spec missed a real need.

A	Adjacent category	Unit tests passed; the gap is in requirements.
B	Camouflage	Headcount does not decide the failure type.
C	Inverted operation	It passed every spec-based test.
D	Correct	Validation checks real user needs.

7. Answer D Concept 4 · transfer

In the V-model, which test level checks the architectural (high-level) design?

Target word: V-model pairing — reversed **Decides:** Architecture pairs with integration.

A	Inverted operation	Pairs with user requirements at the top of the V.
B	Adjacent category	Pairs with detailed module design.
C	Confusion set	Pairs with the system specification.
D	Correct	Checks that components defined in the architecture work together.

8. Answer A Concept 5

At a fintech company ~~with three Scrum teams~~, a senior developer and the Scrum Master disagree about whether feature A or feature B should come first in the product backlog. ~~The developer has 12 years of experience.~~ Who decides?

Target word: role accountability **Decides:** Backlog ordering.

A	Correct	The Product Owner is accountable for backlog order.
B	Camouflage	Experience does not assign accountability.
C	Confusion set	Coaches the process; does not set priorities.
D	Absolute language	Input is welcome, but the Product Owner decides.

9. Answer B Concept 6

A contractor is starting a first-of-its-kind system with major unknowns in its sensor technology. Each cycle must identify the biggest risks and reduce them with prototypes before committing further. ~~The budget is \$40 million.~~ Which model fits?

Target word: process model **Decides:** Risk-driven cycles with prototypes.

A	Dropped element	Drops risk reduction entirely.
B	Correct	Each loop is organized around risk analysis and reduction.
C	Confusion set	Iterative but not organized around explicit risk analysis.
D	Adjacent category	Commits to full requirements before unknowns are resolved.

10. Answer C Concept 7

To make a trade-show date, a team hard-coded prices and skipped tests in the checkout module. Six months later, every change there takes three times as long. ~~The trade show drew 5,000 visitors.~~ What is this, and how should it be handled?

Target word: name and manage **Decides:** Shortcuts now slow every change.

A	Absolute language	High risk; targeted refactoring usually works better.
B	Confusion set	Scope creep is unplanned new work, not shortcuts.
C	Correct	Pay the debt down deliberately.
D	Camouflage	Echoes the event, not the code problem.

11. Answer C Concept 8

A late project has 5 developers. The manager adds 5 more, ~~all strong coders from a top training program.~~ How many communication paths are there now, compared with before?

Target word: communication paths **Decides:** $n(n - 1)/2$.

A	Confusion set	$n(n - 1)$ without dividing by 2.
B	Adjacent category	Uses n^2 .
C	Correct	$10 \times 9 / 2 = 45$; $5 \times 4 / 2 = 10$.
D	Dropped element	Counts people, not paths.

12. Answer C Concept 8 · transfer

A team has 28 communication paths. How many people are on the team?

Target word: solve for n **Decides:** $n(n - 1)/2 = 28$.

A	Confusion set	7 people give 21 paths.
B	Inverted operation	Divides paths by 2.
C	Correct	$8 \times 7 / 2 = 28$.
D	Wrong stem number	The path count.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Choosing Waterfall vs. Agile	Plan-driven Waterfall suits stable, well-defined, heavily regulated requirements; Agile suits requirements that change with feedback.	Treating Agile as always better, or size as the deciding factor.	Regulator-fixed requirements vs. weekly user feedback.
2	Cost of change rises late	A defect costs far more to fix the later it is found; a common rule of thumb is about 10× per phase.	Assuming fix cost grows by a fixed amount per phase.	A fix cost in an early phase and “if found in testing.”
3	Velocity-based planning	Use average velocity to forecast: sprints $\approx$ remaining points / average velocity; capacity = velocity $\times$ sprints.	Planning from the best or worst sprint.	Several sprint point totals and a backlog size.
4	Verification vs. validation (V-model)	Verification: did we build it right (to spec)? Validation: did we build the right thing (meets real needs)? In the V-model, unit $\leftrightarrow$ detailed design, integration $\leftrightarrow$ architecture, system $\leftrightarrow$ system spec, acceptance $\leftrightarrow$ requirements.	Calling a spec-conforming but useless product a verification failure.	Passes tests but users say it does not do what they need.
5	Scrum roles	The Product Owner orders the product backlog; the Scrum Master coaches the process; developers decide how to build.	Letting the Scrum Master or senior developer set priorities.	A dispute over backlog order.
6	Spiral model	The spiral model is risk-driven: each cycle identifies the biggest risks and reduces them, often with prototypes, before committing.	Picking Waterfall for high-unknown projects.	“Major unknowns” and “reduce risks with prototypes.”
7	Technical debt	Shortcuts taken to ship sooner create debt that slows later change; pay it down deliberately through refactoring and tests.	Ignoring it, or rewriting everything at once.	Changes in one area now take much longer after a rushed release.
8	Communication paths (Brooks’s law)	A team of n people has $n(n - 1)/2$ communication paths; adding people to a late project raises overhead and can make it later.	Counting people instead of paths, or using n^2 .	Team size before and after adding staff.

Course scope (public description): Advanced Software Engineering delves into the intricate processes and methodologies essential for developing sophisticated, scalable software architectures. This course offers an in-depth exploration of the software engineering lifecycle, emphasizing a disciplined approach to navigating the complex challenges of software engineering. Students will explore advanced process frameworks and methodologies, including the Waterfall Model and Agile Development, tailored to large-scale and high-stakes projects.